Homework 02: Evaluation of Classification Metrics

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Introduction

The purpose of this analysis was to evaluate a previously scored binary classifier using supplied actual labels, predicted labels, and predicted probabilities. The goal is to calculate and interpret threshold-based classification metrics and probability-based ROC/AUC, then verify the calculations with scikit-learn.

The dataset contained 181 observations, with 3 key columns. The definitions of each column are:

- **class:** the actual class for the observation
- **scored.class**: the predicted class for the observation (based on a threshold of 0.5)
- **scored.probability**: the predicted probability of success for the observation

The analysis used a confusion matrix to calculate classification metrics and an ROC curve to evaluate the model's ability to distinguish between positive and negative cases. The metrics were then calculated manually and then compared with the corresponding functions from scikit-learn to verify the results.

Methods

The actual class labels were compared with the scored.class (predicted class) labels to create a confusion matrix. Class 1 was treated as the positive class, and class 0 was treated as the negative class.

Functions were written to calculate the classification metrics. These metrics calculated include:

- **Accuracy:** The proportion of all observations classified correctly
- **Classification error rate:** The proportion of observations classified incorrectly
- **Precision:** The proportion of predicted positive cases that were actually positive
- **Sensitivity (recall):** The proportion of actual positive cases correctly identified
- **Specificity:** The proportion of actual negative cases correctly identified

- **F1 score:** The harmonic mean of precision and sensitivity
- **Area under the ROC curve (AUC):** A measure of the model's ability to distinguish between positive and negative cases across different classification thresholds

The ROC curve was generated by applying different thresholds to the predicted probabilities and calculating sensitivity and the false positive rate at each threshold. Youden's J statistic was also used to identify a threshold that balances sensitivity and specificity.

Finally, the results were compared with scikit-learn to verify the calculations.

Results

Confusion Matrix

The confusion matrix produced the following results:

Metric	Count	Percentage	Interpretation
True Negatives	119	65.75%	The actual class was 0 and the model correctly predicted 0
False Positives	5	2.76%	The actual class was 0 and the model correctly predicted 1 (type I error)
False Negatives	30	16.57%	The actual class was 1 and the model correctly predicted 0 (type II error)
True Positives	27	14.92%	The actual class was 1 and the model correctly predicted 1

The model correctly classified 119 negative cases and 27 positive cases. It incorrectly classified 5 negative cases as positive and 30 positive cases as negative.

Classification Metrics

The calculated classification metrics were:

Metric	Result
Accuracy	0.8066
Classification error rate	0.1934
Precision	0.8438
Sensitivity	0.4737
Specificity	0.9597
F1 score	0.6067
AUC	0.8503

The model achieved an accuracy of 80.66%, meaning that it correctly classified approximately 81% of the observations. Precision was 84.38%, indicating that most cases predicted as positive were actually positive. Specificity was 95.97%, showing that the model correctly identified most actual negative cases. In contrast, sensitivity was 47.37%, meaning that the model identified fewer than half of the actual positive cases. The 30 false negatives compared with only 5 false positives further demonstrate this limitation.

The F1 score of 0.6067 shows the balance between precision and sensitivity. Although the model performed well in identifying negative cases and making positive predictions, its relatively low sensitivity reduced the overall F1 score.

ROC Curve and AUC

The ROC curve was used to evaluate the classifier across different probability thresholds. The AUC was approximately 0.849 using the 0.01 threshold grid and 0.850 using all distinct probability thresholds. An AUC of 0.85 indicates that the model can distinguish between positive and negative cases relatively well. The ROC curve rises steeply at low false positive rates, suggesting that lowering the threshold can identify more positive cases while initially adding relatively few false positives.

At the default threshold of 0.5, sensitivity is 0.474 and specificity is 0.960, meaning the model identifies fewer than half of the actual positive cases. Using Youden's J statistic, a threshold of 0.37 provides a more balanced result, increasing sensitivity to approximately 0.70 while reducing specificity to approximately 0.85. The preferred threshold depends on the relative costs of false negatives and false positives. However, because the threshold was selected and evaluated using the same 181 observations, its performance may be optimistic when applied to new data.

Verification and evaluation with scikit-learn

The manually calculated metrics were compared with the corresponding scikit-learn functions. The results are shown below.

Metric	Manual	Scikit-learn
Accuracy	0.8066	0.8066
Precision	0.8438	0.8438
Sensitivity	0.4737	0.4737
Specificity	0.9597	0.9597
F1 score	0.6067	0.6067
AUC	0.8503	0.8503

All six metrics matched between the manual calculations and scikit-learn. This confirms that the manual implementations produced results consistent with the corresponding library functions.

Implementing the metrics manually also helped demonstrate how each measure is calculated from the confusion matrix and predicted probabilities. Scikit-learn provided an independent way to verify the calculations.

Discussion

The results show that the classifier's performance depends on which metric is considered. While accuracy was relatively high, the model missed more than half of the actual positive cases. This highlights a limitation of relying on accuracy alone when evaluating a classifier.

The dataset contained 124 negative cases out of 181 observations, or approximately 68.5%. A classifier that always predicted the negative class would therefore achieve 68.5% accuracy. The model's accuracy of 80.66% is higher than this baseline, but its sensitivity indicates that identifying positive cases remains a challenge.

The AUC of 0.8503 suggests that the model's predicted probabilities have good discriminatory ability, even though its sensitivity at the default threshold is relatively low. This distinction shows why examining the ROC curve and considering alternative thresholds can be useful. Any threshold adjustment should account for the trade-off between identifying more positive cases and generating additional false positives.

Conclusion

This analysis evaluated various classification metrics in python. The manual calculations matched the corresponding scikit-learn results for all six evaluated metrics, confirming the consistency of both methods.

The model achieved 80.66% accuracy and an AUC of 0.8503, suggesting reasonable overall classification performance and good discriminatory ability. However, its sensitivity of 47.37% indicates that it missed a substantial proportion of positive cases. Evaluating alternative classification thresholds may improve sensitivity.

Overall, the analysis demonstrates the importance of using multiple metrics to evaluate a model and understanding how these metrics are calculated.

Appendix

Source code: https://github.com/alinsimon/data621-data-mining/blob/main/homeworks/hw02/notebooks/hw02_classification_metrics_Final.ipynb